# Elham Estiri

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## Summary

PhD candidate in Mechatronics Engineering (Kent State, defense targeted Fall 2027) working on Bayesian deep learning and state-space modeling for safety-critical physiological systems. Builds models that combine Bayesian variational autoencoders with Unscented Kalman Filtering to estimate hidden states and quantify aleatoric and epistemic uncertainty, and tests how they generalize to out-of-distribution data (8 peer-reviewed papers). Background in linear algebra, optimization, and stochastic processes. Interested in bringing this probabilistic perspective to continual learning in deep networks.

## Skills

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| **Programming Languages** | Python (Expert), CUDA, C++, MATLAB, SQL |
| **Probabilistic & Causal ML** | causal state-space modeling, Unscented Kalman Filtering (UKF), constrained ELBO optimization, uncertainty quantification (aleatoric & epistemic) |
| **ML & Deep Learning** | PyTorch, TensorFlow, Transformer architectures, Hugging Face, reinforcement learning (RL), Bayesian deep learning (VAEs, BNNs), supervised learning, model evaluation |
| **Data Science & Math** | NumPy, Pandas, statistical modeling, linear algebra, optimization, stochastic processes |
| **Research Methodology** | experimental design, statistical hypothesis testing, OOD generalization testing, peer review |
| **ML Systems & Inference** | LLM inference optimization, model quantization, model pruning, custom CUDA kernel development, GPU profiling (Nsight, nvprof) |
| **Software Engineering** | Git, Docker, Linux, code review, reproducible ML pipelines, unit testing, Claude Code, Codex |

## Experience

**Graduate Research Assistant** **2022 – Present**

Kent State University, College of Aeronautics and Engineering, Dept. of Mechanical Engineering, Kent, OH

- Developed causal deep state-space models in PyTorch combined with a Bayesian VAE and Unscented Kalman Filter framework ("UKF-BVAE"), integrating aleatoric and epistemic uncertainty quantification for risk-aware, interpretable decision-making in a safety-critical physiological control system.
- Formulated sequential latent-state inference as a constrained ELBO optimization problem; validated on preclinical physiological data and extending to the MIMIC-IV clinical database and an independent dataset to test out-of-distribution generalization.
- Applied Bayesian VAEs to out-of-distribution detection in physiological modeling (IEEE EMBC 2024), and designed a Q-learning RL policy benchmarked against PID and rule-based controllers.
- Published 8 peer-reviewed papers (venues include ACC, IEEE EMBC, IEEE Access, ASME JDSMC) on Bayesian/causal modeling, uncertainty quantification, and control for safety-critical systems.

## Projects

**On-Device LLM Inference Pipeline (ModularMAX)** GitHub/PR

- Built a native inference pipeline for a family of generative language models (270M–3B parameters) in an open-source graph-based inference engine, measuring a 52% throughput increase over baseline; verified output against Hugging Face reference checkpoints with a 27-test suite.

**Efficient Transformer Optimization and Deployment** GitHub

- Implemented a fused-softmax operation as a custom CUDA kernel after finding no speedup in pure PyTorch; verified correctness and benchmarked against CPU and native GPU baselines.
- Benchmarked quantization and pruning against a PyTorch transformer baseline on the full SST-2 validation set; found and corrected a sampling-bias flaw in an earlier evaluation.

## Education

**Ph.D., Mechatronics Engineering** | *Kent State University, Kent, OH* 2022 – Present (expected Fall 2027)

**M.S., Biomedical Engineering** | *Hakim Sabzevari University, Iran* 2018 – 2020

**B.S., Electrical Engineering** | *Hamedan University of Technology, Iran* 2013 – 2017

## Selected Publications (full list on Google Scholar)


- E. Estiri, H. Mirinejad. Adaptive Uncertainty-Aware Modeling and Stochastic Radial Basis Function Predictive Control for Personalized Fluid Resuscitation. *IEEE Trans. Control Systems Technology (TCST)*, accepted, 2026; arXiv:2609.21821.
- E. Estiri, H. Mirinejad. Bayesian Variational Autoencoders for Out-of-Distribution Detection in Physiological Modeling: A Case Study in Fluid Therapy. *IEEE EMBC*, 2024.
- E. Estiri, H. Mirinejad. Bayesian Nonlinear State-Space Modeling and Chance-Constrained Predictive Control for Hemodynamic Resuscitation. *2026 American Control Conference (ACC)*, pp. 1922–1927, 2026.
- E. Estiri, H. Mirinejad. Automated Fluid Resuscitation Via Robust Nonlinear State-Space Modeling and Radial Basis Function Optimal Control. *Journal of Dynamic Systems, Measurement, and Control*, 148(6), 061001, 2026.
- E. Estiri, H. Mirinejad. Closed-Loop Control of Fluid Resuscitation Using Reinforcement Learning. *IEEE Access*, 2023.